Diary Data Analysis

Anne Gaule

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1) Group numbers for analyses:

- Main analyses
  - Child psb = 83
  - Teach psb = 102

Table 1: Child PSB

Training_Condition	n
CONTROL	44
TRAIN	39

```
## [1] "n participants total = "
```

```
## [1] 83
```

Table 2: teach PSB

Training_Condition	n
CONTROL	53
TRAIN	49

```
## [1] "n participants total = "
```

```
## [1] 102
```

2) Impact of Training on Child Rated Prosocial Behaviour at Two Weeks

2.1) Plotting Child Prosocial Behaviour Diary Ratings at D1 & D2



Difference plot:



2.2) Model Specification in Preregistration

Model specification in SAP:

$$\text{Child_PSB}_j = (\beta_0 + u_k) + \beta_1 \text{PSB.0}_j + \beta_2 \text{Train}_j + \varepsilon_j$$

Where:

- Child_PSB_j: Child-rated antisocial behaviour at D2 for participant j.
- $j = 1, \ldots, N$ identifies the participants.
- β_0 is a fixed intercept representing the average baseline level of the dependent variable (Child_PSB_j) across all participants when all predictors are zero.
- $u_k \sim \mathcal{N}(0, \sigma_u^2)$ is the random effect for schools, accounting for variability between schools.
- $\beta_1 \text{PSB.0}_j$: effect of child-rated antisocial behaviour score at baseline (D1) on ASB at D2.
- β_2 is the fixed effect of training condition (Train; an indicator variable coded as 0 for controls or 1 for training group) on the outcome, indicating how being in the training group influences the outcome.
- $\varepsilon_j \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ is random error.

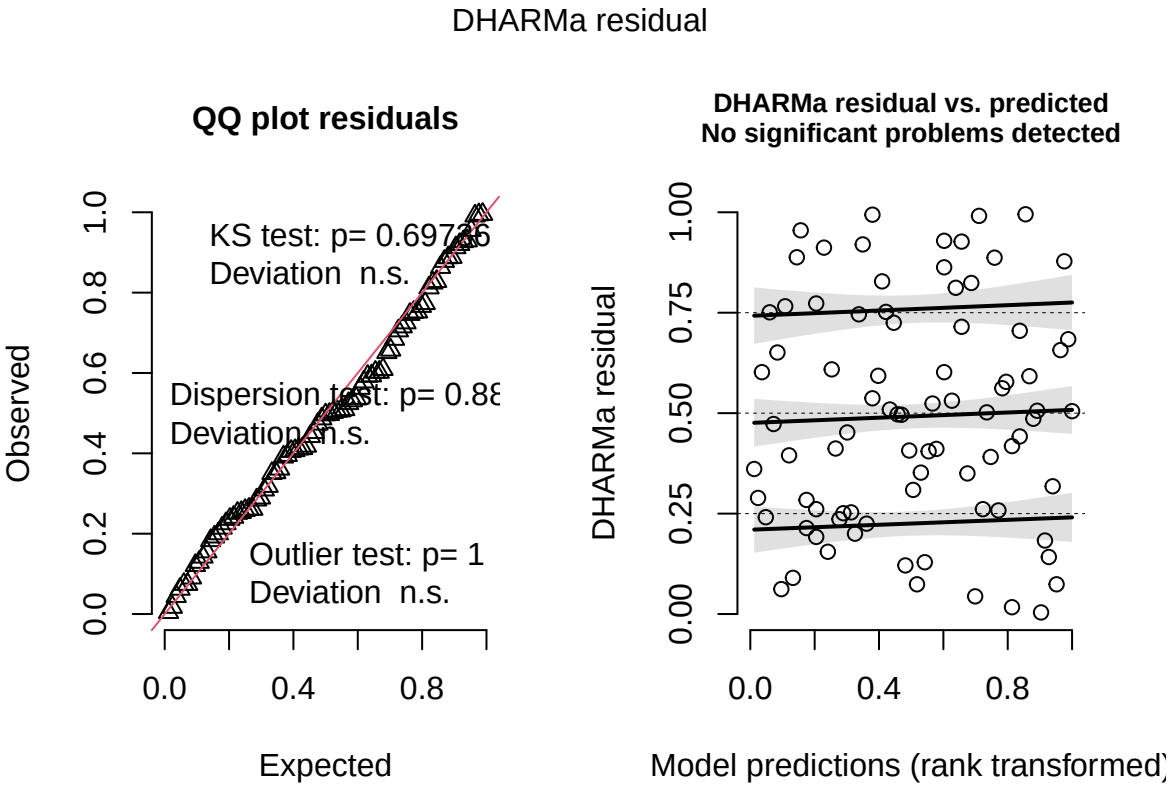
2.3) Preregistered Model: Linear mixed model predicting child Prosocial behaviour at D2 by training condition, adjusted for behaviour at D1

This model does not give a singular fit error. However, the bootstrapped version (run for consistency with other models) does.

```
model_c_psb_1 <- lmer(PSB_sum_D2 ~ PSB_sum_D1 + Training_Condition +
                      (1|School.ID),
                      data =  child_psb_d1d2_wide)
summary(model_c_psb_1)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: PSB_sum_D2 ~ PSB_sum_D1 + Training_Condition + (1 | School.ID)
##    Data: child_psb_d1d2_wide
##
## REML criterion at convergence: 705.6
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.63962 -0.64724 -0.06347  0.56513  2.62439
##
## Random effects:
##  Groups    Name        Variance Std.Dev.
##  School.ID (Intercept)   8.25    2.872
##  Residual              309.74   17.599
## Number of obs: 83, groups:  School.ID, 15
##
## Fixed effects:
##                          Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)                0.2614     5.7139 70.8656   0.046    0.964
## PSB_sum_D1                 0.9873     0.0932 79.9066  10.593   <2e-16 ***
## Training_ConditionTRAIN    5.9619     3.9365 76.5588   1.514    0.134
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) PSB__D
## PSB_sum_D1  -0.871
## Trnn_CTRAIN -0.451  0.155
```

2.3.1) Checking Residuals:



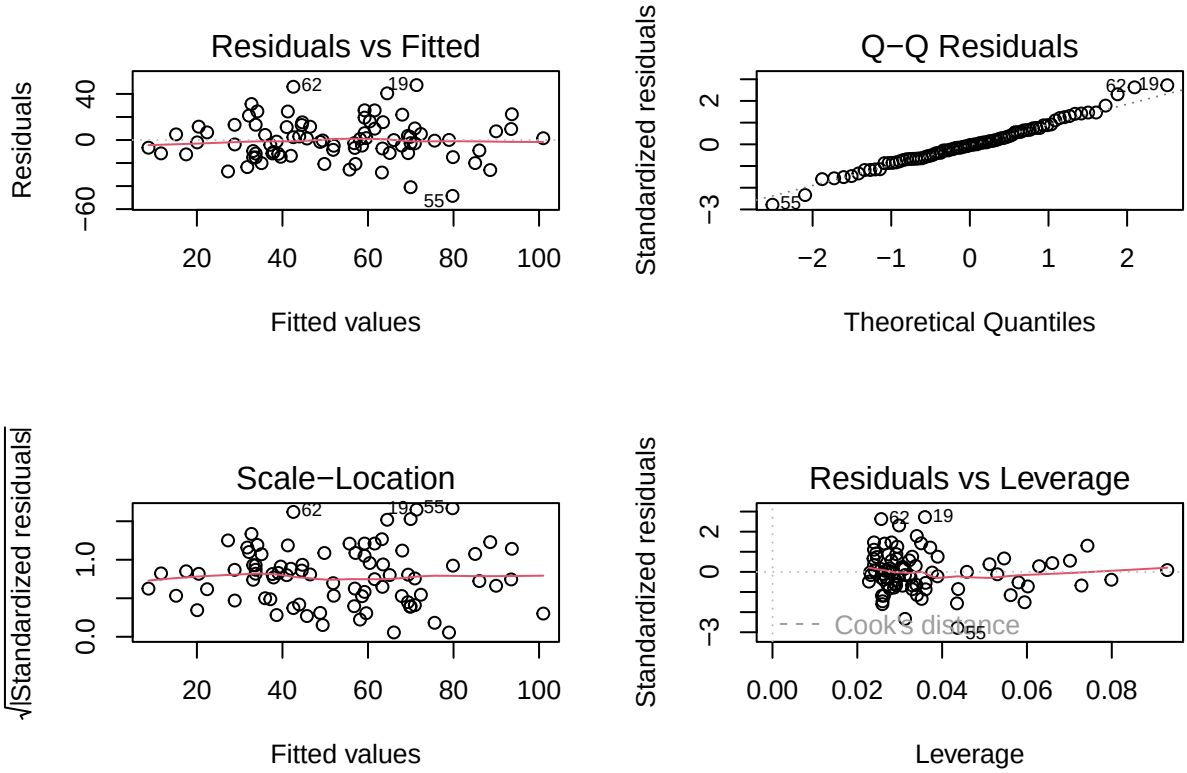
2.4) Bootstrapping Preregistered LMM: This gives singular fit error

```
# REB bootstrap advised when assumptions are violated
boot_results <- lmeresampler::bootstrap(model_c_psb_1, .f = fixef, type = "reb", B = 5000)
```

```
## Warning in reb_bootstrap.lmerMod(model, .f, B, reb_type, .refit): 'reb_type'
## unspecified, performing REB 0 bootstrap
```

2.4) Running simple regression instead

2.4.1) Plotting Residuals



2.5) Final Model: Bootstrapped Linear Regression

```
# non-parametric case bootstrap: https://journal.r-project.org/articles/RN-2002-017/RN-2002-017.pdf
boot_coef_c_psb <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(PSB_sum_D2 ~ PSB_sum_D1 + Training_Condition,
                    data =  d)
  return(coef(fit))
}


# Run bootstrap
set.seed(123)  # For reproducibility
boot_results_c_psb <- boot(data = child_psb_d1d2_wide,
                      statistic = boot_coef_c_psb,
                      R = 5000)
```

2.5.1) Results

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = child_psb_d1d2_wide, statistic = boot_coef_c_psb,
##     R = 5000)
##
##
## Bootstrap Statistics :
##      original       bias    std. error
## t1* 0.4416934  0.079490022  4.96672250
## t2* 0.9793064 -0.002067222  0.08847852
## t3* 6.2717871  0.052609594  3.89891917
```

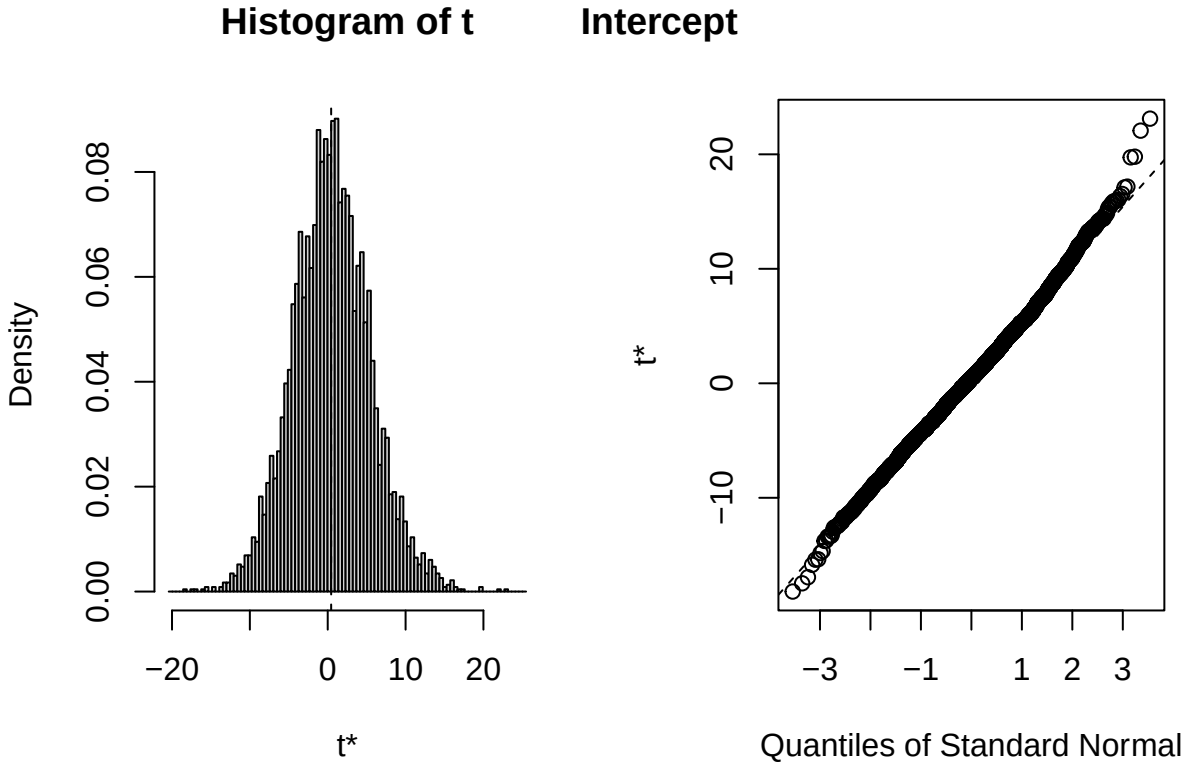
2.5.2) Confidence Intervals

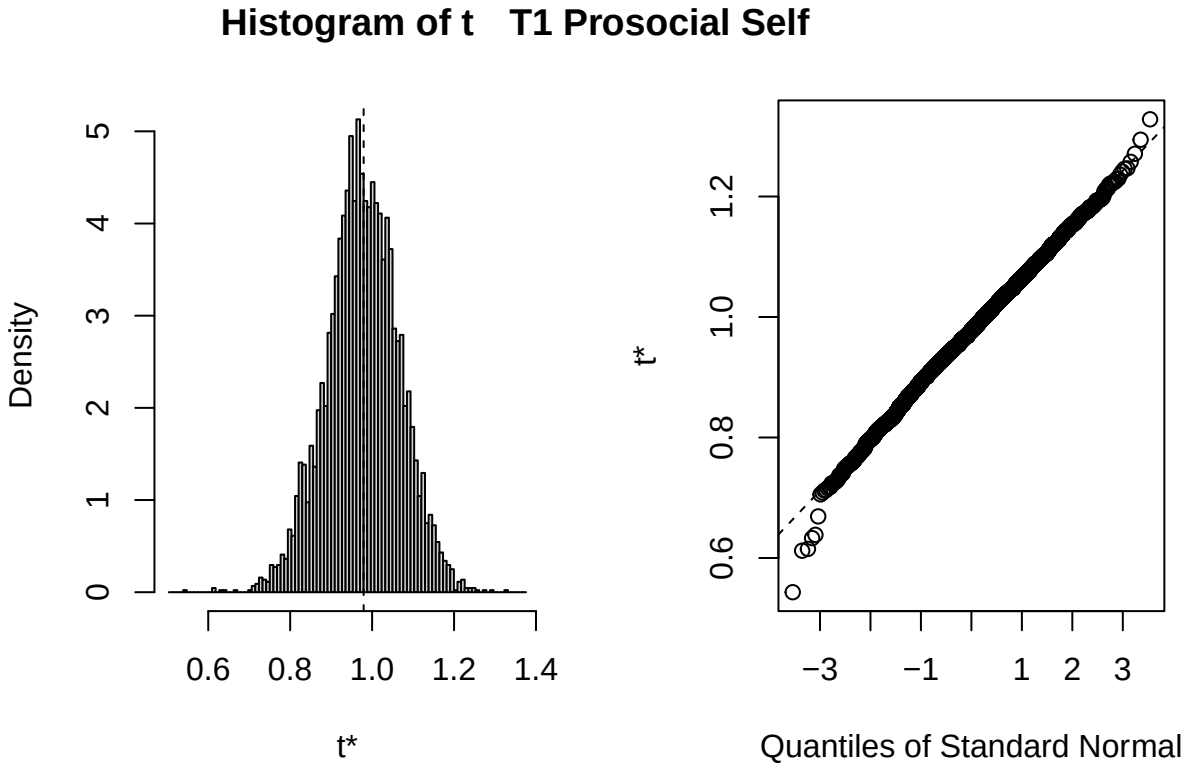
```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_psb, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   (-8.9081, 10.8790 )
## Calculations and Intervals on Original Scale

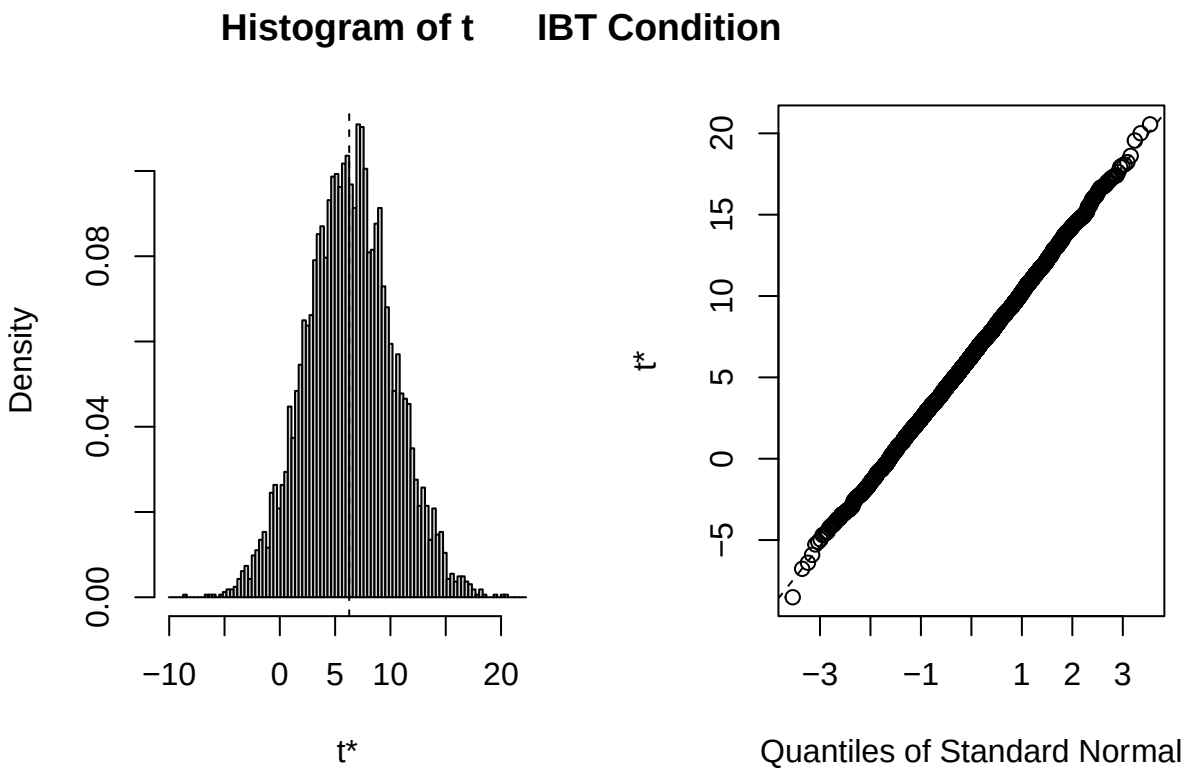
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_psb, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.8011,  1.1518 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_psb, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-1.250, 14.217 )
## Calculations and Intervals on Original Scale
```

2.5.3) Bootstrap Diagnostics

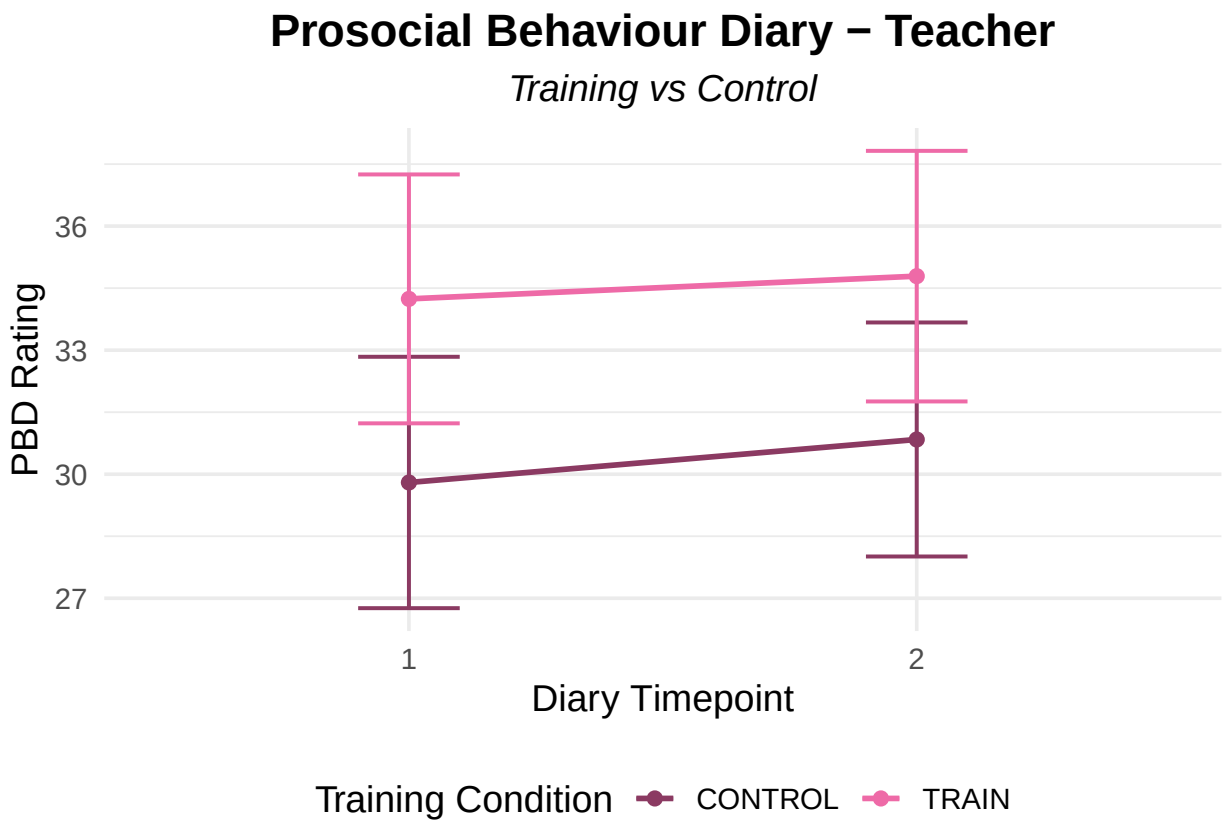





Histogram of t IBT Condition
Density
0.00
0.04
0.08
−10
0
5
10
20
t*
t*
−5
0
5
10
15
20
−3
−1
1
2
3
Quantiles of Standard Normal

3) Impact of Training on Teacher Rated Prosocial Behaviour at Two Weeks

3.1) Plotting teacher rated Prosocial Behaviour at D1 & D2



Difference plot:



2.2) Model Specification in preregistration

$$\text{Teach_PSB}_j = (\beta_0 + u_k + u_{i(k)}) + \beta_1 \text{PSB.0}_j + \beta_2 \text{Train}_j + \varepsilon_j$$

where:

- Teach_PSB_j: Teacher-rated prosocial behaviour (at D2) for participant j.
- $j = 1, \ldots, N$ identifies the participants.
- $u_k \sim \mathcal{N}(0, \sigma_u^2)$ is the random effect for schools, accounting for variability between schools.
- $u_{i(k)} \sim \mathcal{N}(0, \sigma_u^2)$ is the random effect for teachers, nested within schools.
- β_0 is a fixed intercept representing the average baseline level of the dependent variable (Teach_ASB_j) across all participants when all predictors are zero.
- $\beta_1 \text{PSB.0}_j$: effect of teacher-rated antisocial behaviour score at baseline (D1) on later prosocial behaviour.
- β_2 is the fixed effect of Training condition (Train; an indicator variable coded as 0 for controls or 1 for training group) on the outcome, indicating how being in the training group influences the outcome.
- $\varepsilon_j \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ is the random error term.

2.3) Preregistered Linear mixed model predicting ASB at D2 by training condition, adjusted for behaviour at D1.

Once again model has a singular fit. This time it seems to be school that is not contributing to the model.

```
model_t_psb_1 <- lmer(PSB_mean_D2~ PSB_mean_D1 + Training_Condition +
                      (1|School.ID/teachUUID_diaries),
                      data =  teach_psb_d1d2_wide)
```

```
## boundary (singular) fit: see help('isSingular')
```

2.3.1) Investigating singular fit

```
##  Groups                       Name        Std.Dev.
##  teachUUID_diaries:School.ID (Intercept)  8.5374
##  School.ID                   (Intercept)  0.0000
##  Residual                                 12.8362
```

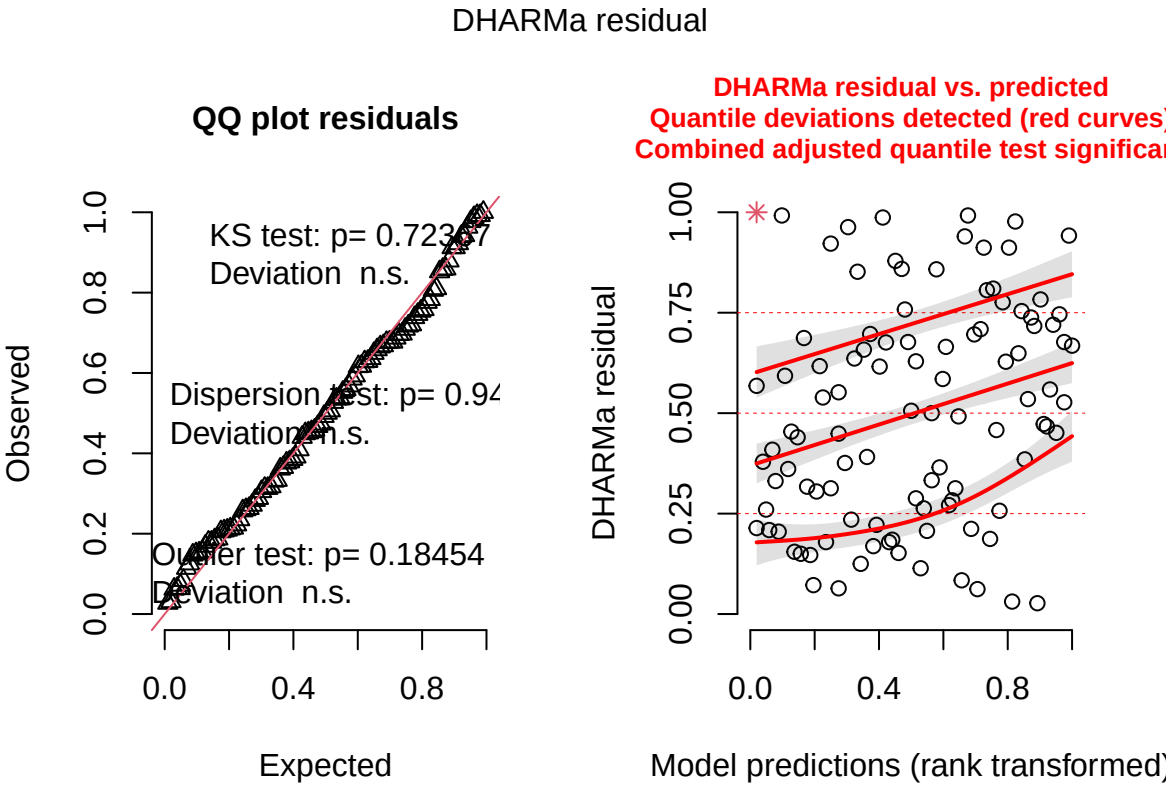
2.4) Simplified LMM with random effect of teacher only. This does fit.

```
model_t_psb_2 <- lmer(PSB_mean_D2~ PSB_mean_D1 + Training_Condition +
                      (1|teachUUID_diaries),
                      data =  teach_psb_d1d2_wide)

summary(model_t_psb_2)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## PSB_mean_D2 ~ PSB_mean_D1 + Training_Condition + (1 | teachUUID_diaries)
##    Data: teach_psb_d1d2_wide
##
## REML criterion at convergence: 830.3
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.5397 -0.5454 -0.1350  0.5934  3.7000
##
## Random effects:
##  Groups            Name        Variance Std.Dev.
##  teachUUID_diaries (Intercept)  76.25    8.732
##  Residual                      163.34   12.780
## Number of obs: 102, groups:  teachUUID_diaries, 36
##
## Fixed effects:
##                          Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)              12.83639    3.09559 80.80741   4.147 8.27e-05 ***
## PSB_mean_D1               0.56481    0.07311 98.18077   7.725 9.64e-12 ***
## Training_ConditionTRAIN  2.09281    2.85929 93.31834   0.732    0.466
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) PSB__D
## PSB_mean_D1 -0.622
## Trnn_CTRAIN -0.304 -0.184
```

2.4.1) Assumption checks show serious violations



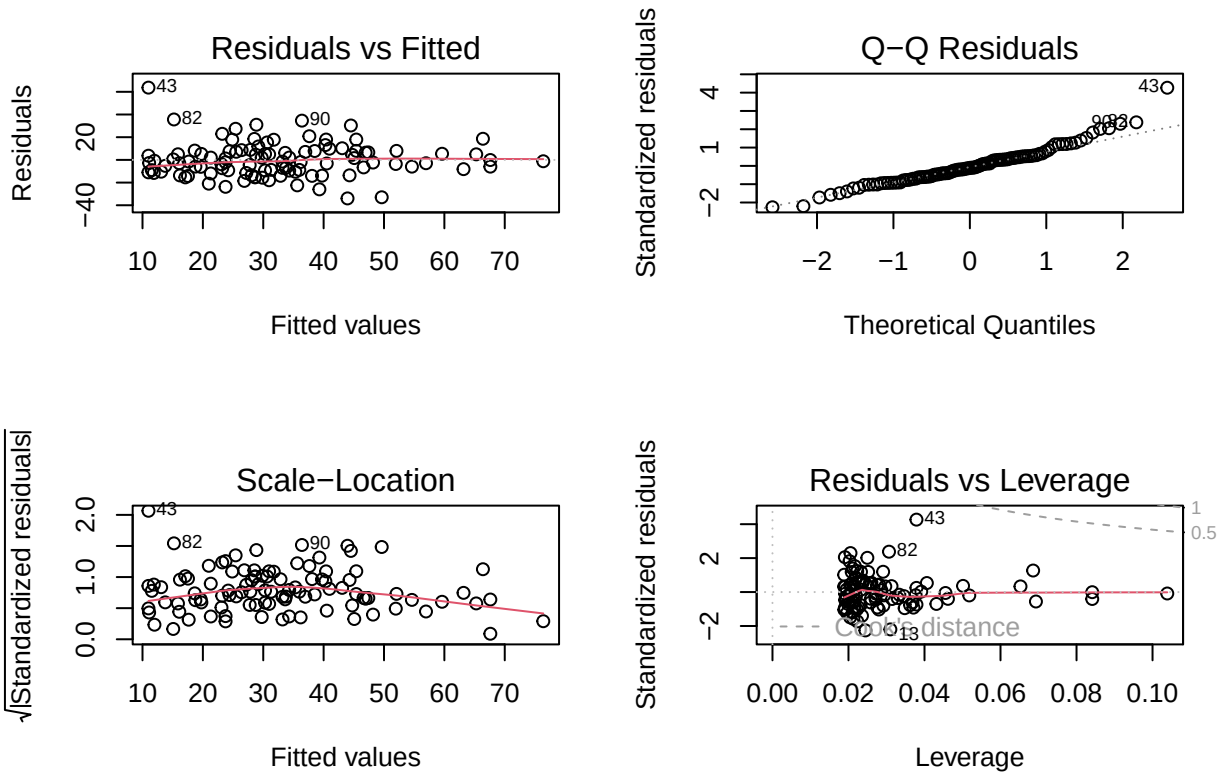
2.5) Simple Linear Regression

```
model_t_psb_2_lm <- lm(PSB_mean_D2~ PSB_mean_D1 + Training_Condition ,
                       data =  teach_psb_d1d2_wide)

summary(model_t_psb_2_lm)
```

```
##
## Call:
## lm(formula = PSB_mean_D2 ~ PSB_mean_D1 + Training_Condition,
##     data = teach_psb_d1d2_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -33.958  -9.424  -1.693   7.472  63.679
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)              10.98796    2.96116   3.711 0.000341 ***
## PSB_mean_D1               0.66619    0.07041   9.462 1.63e-15 ***
## Training_ConditionTRAIN   0.99317    3.03112   0.328 0.743862
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.21 on 99 degrees of freedom
## Multiple R-squared:  0.4796, Adjusted R-squared:  0.4691
## F-statistic: 45.62 on 2 and 99 DF,  p-value: 9.068e-15
```

2.5.1) Assumption Checks (indicate LM better fit)



2.6) Final Model: Bootstrapped LM model

```
# non-parametric case bootstrap: https://journal.r-project.org/articles/RN-2002-017/RN-2002-017.pdf
# argued to be better  for heteroskedastic data:
boot_coef_t_psb <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(PSB_mean_D2 ~ PSB_mean_D1 + Training_Condition,
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
boot_results_t_psb <- boot(data = teach_psb_d1d2_wide,
                     statistic = boot_coef_t_psb,
                     R = 5000)
```

2.6.1) Results

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = teach_psb_d1d2_wide, statistic = boot_coef_t_psb,
##     R = 5000)
##
##
## Bootstrap Statistics :
##       original        bias    std. error
## t1* 10.9879561 -0.0523952374  3.54609196
## t2*  0.6661914 -0.0001248583  0.06537699
## t3*  0.9931694  0.0995126731  2.89251590
```

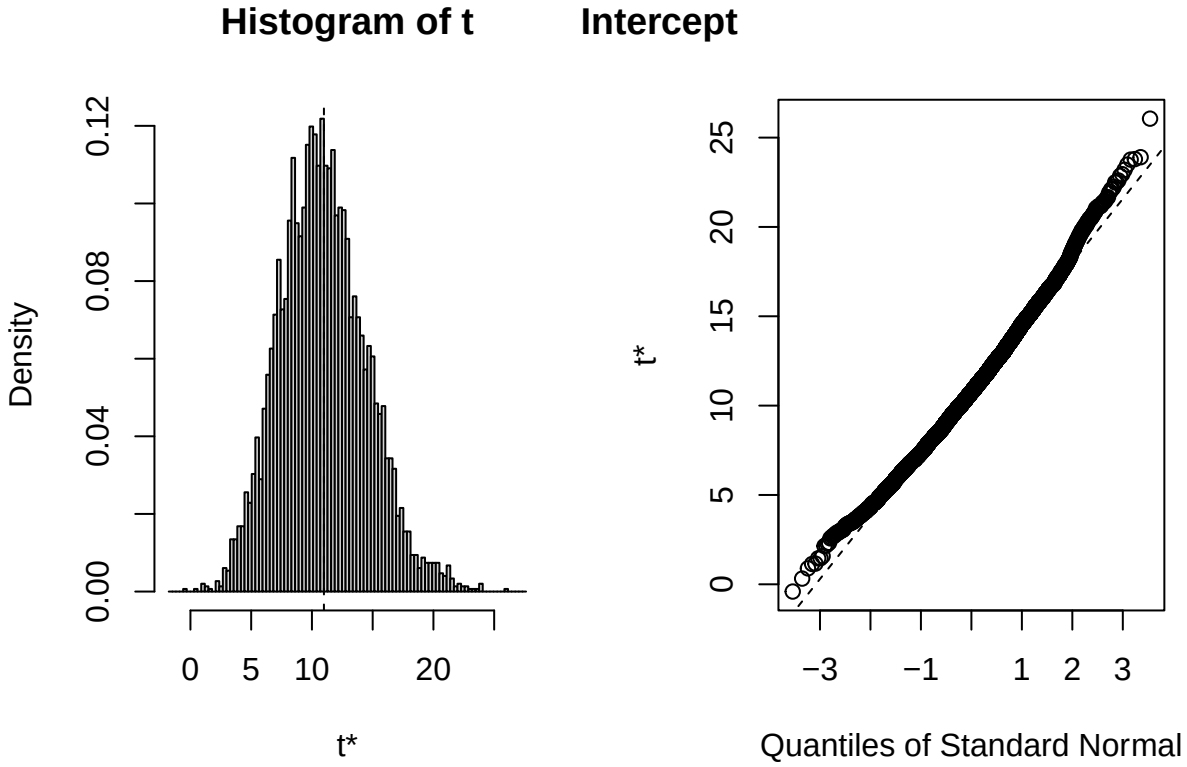
2.6.2) Confidence Intervals

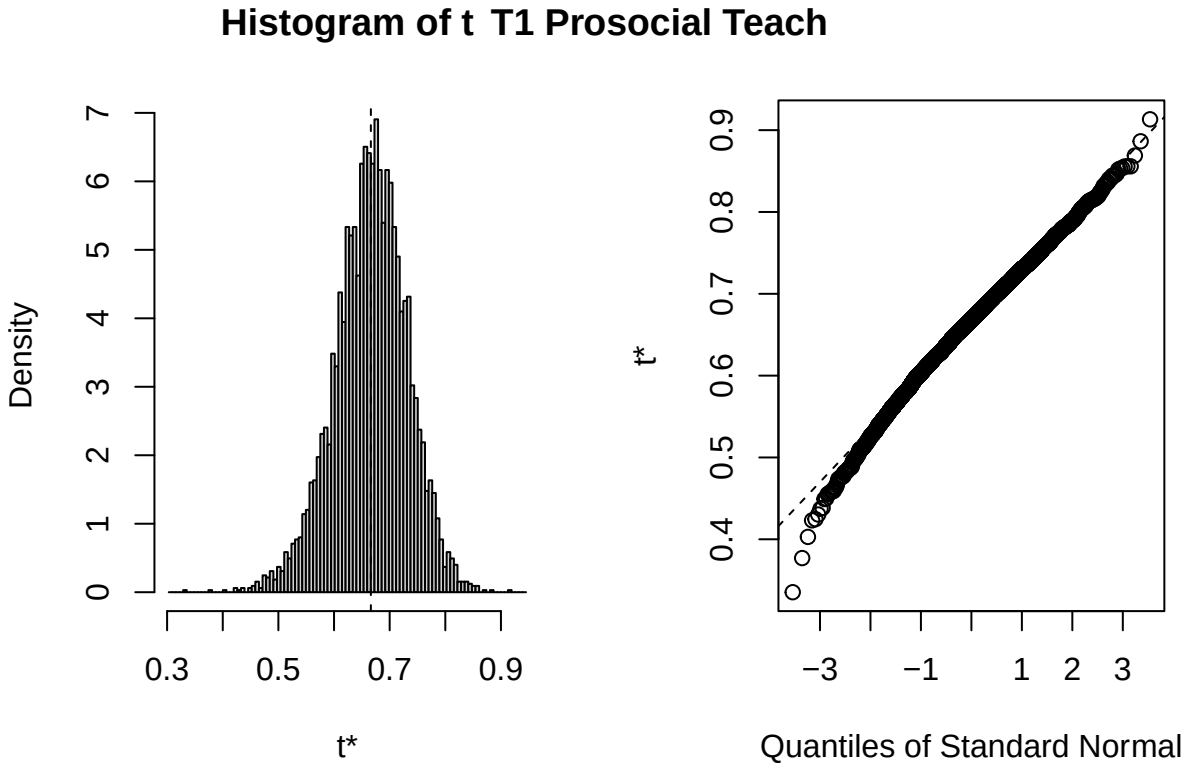
```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_psb, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   ( 5.47, 20.59 )
## Calculations and Intervals on Original Scale
```

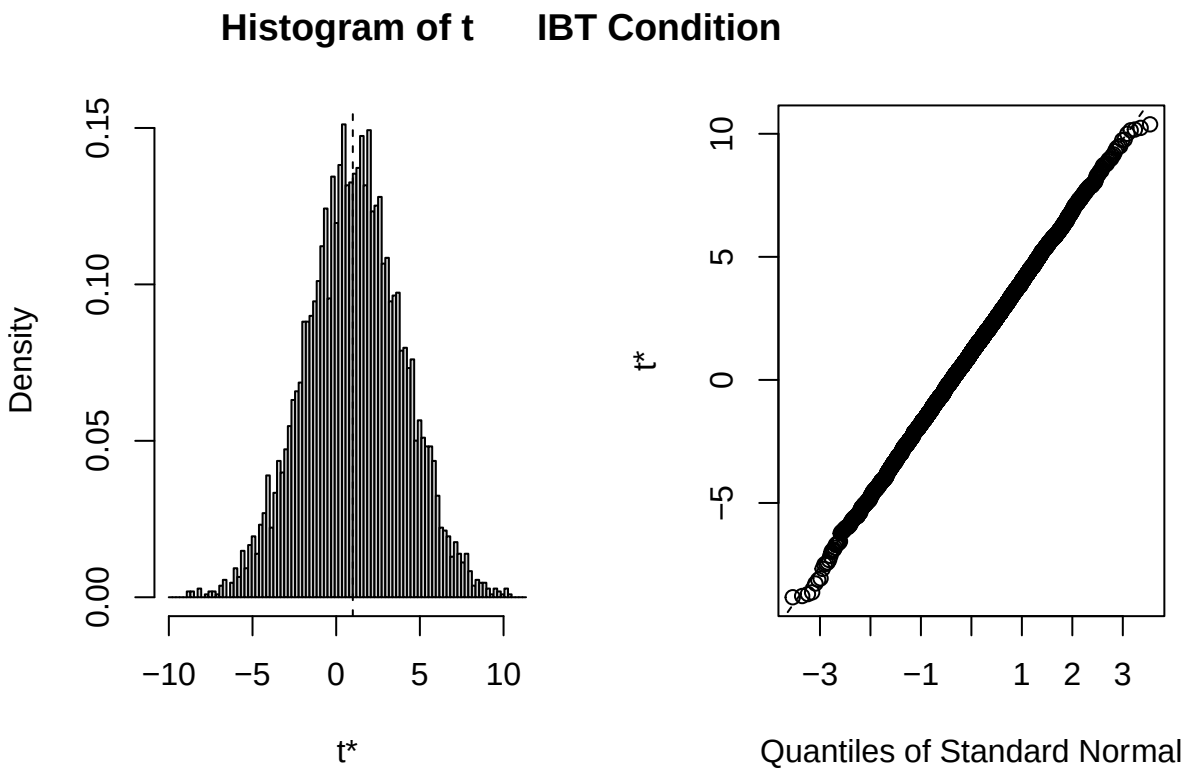
```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_psb, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.5065,  0.7747 )
## Calculations and Intervals on Original Scale
```

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_psb, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-4.9988,  6.3597 )
## Calculations and Intervals on Original Scale
```

2.6.3) Bootstrap Diagnostics






Histogram of t IBT Condition
Density
0.00
0.05
0.10
0.15
−10
−5
0
5
10
t*
t*
−5
0
5
10
−3
−1
1
2
3
Quantiles of Standard Normal